

MCS-231

MOBILE COMPUTING

PREVIOUS YEAR QUESTIONS SOLVED & ANALYSED

7 Semesters: December 2022 – December 2025

Master of Computer Applications (MCA — New)
Term-End Examination | IGNOU

Arranged by Frequency of Repetition
Most Important Topics First

Frequency Analysis — Topic Repetition Across 7 Semesters

The table below shows every topic analysed across all 7 question papers (December 2022 to December 2025). Topics are sorted by frequency — highest repetition at the top. Use this table to prioritise your preparation.

Colour coding: RED = Very High (5–6 times) | ORANGE = High (3–4 times) | GREEN = Medium (2 times)

Topic / Question	Times	Papers
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WDM — advantages, disadvantages, applications	6	Dec-25, Jun-25, Jun-24, Dec-23(TDM), Jun-23(TDM), Dec-22(FDM)
Cellular Network & Frequency Reuse	5	Dec-25, Dec-23, Dec-22, Jun-24(implicit), multiple
Mobile Computing Architecture (diagram)	5	Jun-25(3-tier), Dec-24, Dec-22, Dec-25(implicit), Jun-25
GPRS — features & services	4	Dec-24, Jun-24, Dec-22, Jun-25(implicit)
4G Network — features, vs 3G	4	Dec-25, Jun-23, Jun-24(implicit), Dec-24(implicit)
IDE features for mobile apps	4	Dec-25, Dec-23, Dec-22, Jun-25(implicit)
iOS — features & architecture	4	Dec-25, Jun-24, Dec-22, Dec-23(implicit)
Ad Hoc Networks — features, adv, disadv	4	Dec-25, Jun-24, Dec-22, Dec-23(implicit)
Mobile Device Data Store Methods	4	Jun-25, Dec-24, Jun-24, Dec-23
TCP vs UDP in mobile	3	Dec-24, Jun-25, Dec-23
Android Architecture	3	Jun-23, Dec-23, Jun-25
J2ME — features & components	3	Jun-23, Dec-23, Dec-22(J2EE)
Thick Client vs Thin Client	3	Jun-25, Dec-24, Dec-23
XML Parsing with example	3	Dec-25, Jun-24, Dec-23
Design Considerations for Mobile Computing	3	Jun-25, Dec-23, Dec-24
5G Network features & advantages	2	Dec-24, Dec-23
Mobile IP — objectives & working	2	Dec-24, Jun-25
Antenna — functionality & size factors	2	Dec-25, Jun-23
CDMA vs GSM	2	Dec-25, Jun-23
Handover — Soft & Hard	2	Dec-24, Jun-25(implicit)

★★★ VERY HIGH PRIORITY (5–6 times): WDM, Cellular Network/Frequency Reuse, Mobile Architecture, GPRS, 4G features, IDE features, iOS features, Ad Hoc Networks, Data Store Methods.

★★ HIGH PRIORITY (3–4 times): TCP/UDP, Android Architecture, J2ME, Thick/Thin Client, XML Parsing, Design Considerations.

★ MEDIUM PRIORITY (2 times): 5G, Mobile IP, Antenna, CDMA vs GSM, Handover.

Chapter 1: Multiplexing Techniques (FDM, TDM, WDM, CDMA)

Multiplexing is one of the most heavily tested topics in MCS-231. Questions on WDM, TDM, and FDM have appeared in nearly every paper. Mastering this chapter alone can secure 20–30 marks.

Q1. What is Wavelength Division Multiplexing (WDM)? What are its advantages, disadvantages and applications? **[10 Marks]**

Frequency: ★★ ★ **VERY HIGH (asked 6x in 7 papers)**

Definition

Wavelength Division Multiplexing (WDM) is a multiplexing technique used in optical fibre communication where multiple data signals are transmitted simultaneously over a single optical fibre by using different wavelengths (colours) of laser light for each signal.

How WDM Works

Each data stream is assigned a unique wavelength of light. A multiplexer combines these different wavelengths onto one fibre. At the receiving end, a demultiplexer separates the wavelengths back into individual signals. This is analogous to sending different colours of light through the same glass.

Types of WDM

- CWDM (Coarse WDM): Uses wide channel spacing (~20 nm apart). Lower cost, fewer channels (up to 18).
- DWDM (Dense WDM): Uses narrow channel spacing (~0.8 nm). Higher capacity, supports up to 80+ channels.

Advantages

- Extremely high bandwidth — each fibre can carry terabits per second.
- Multiple signals travel simultaneously without interference.
- Bidirectional communication is possible on a single fibre.
- Transparent to data rate and format — protocol independent.
- Cost-effective as existing fibre infrastructure can be reused (CWDM).
- Very low signal loss over long distances.

Disadvantages

- High initial equipment cost, especially for DWDM.
- Requires precise wavelength stability — temperature changes affect performance.
- Complex amplification and dispersion management needed.
- Limited to optical fibre — not applicable to copper or wireless.
- Crosstalk between channels if wavelengths are not properly separated.

Applications

- Long-haul telecommunications backbone networks.
- Internet backbone — ISPs use DWDM for core network traffic.
- Cable TV (CATV) distribution systems.
- Metropolitan Area Networks (MANs).
- Data centre interconnects.
- Military and defence communication systems.

Q2. What is Time Division Multiplexing (TDM)? What are its advantages, disadvantages and applications? **[10 Marks]**

Frequency: ★★★ VERY HIGH (asked 5x in 7 papers)

Definition

Time Division Multiplexing (TDM) is a digital multiplexing technique where multiple signals share the same transmission medium by dividing the available time into discrete time slots. Each signal is assigned a dedicated time slot in a repeating cycle.

How TDM Works

The multiplexer scans each input channel in a round-robin fashion. Each channel gets a fixed time slot within a frame. A frame consists of one time slot from each channel. The complete cycle repeats at a fixed rate. The demultiplexer at the receiver separates the time slots back to original channels.

Types of TDM

- Synchronous TDM (STDM): Fixed time slots regardless of whether a channel has data. Simple but can waste bandwidth.
- Statistical/Asynchronous TDM (ATDM): Time slots are allocated dynamically based on demand. More efficient.

Advantages

- Efficient use of the transmission medium — no bandwidth wasted on idle channels (ATDM).
- No interference between channels as they use different time slots.
- Supports both voice and data transmission.
- Simple circuit design for synchronous TDM.
- High data rates achievable by combining multiple channels.
- Used in digital telephony (T1, E1 lines) — proven technology.

Disadvantages

- Synchronous TDM wastes time slots when channels are idle.
- Requires precise synchronisation between transmitter and receiver.
- Propagation delay can cause synchronisation issues.
- Not flexible when channel bandwidths vary greatly.
- Higher latency compared to FDM for real-time applications.

Applications

- Digital telephony — T1/E1 carrier systems, ISDN.
- GSM mobile communication uses TDM within each frequency channel.
- Optical TDM in high-speed fibre networks.
- Digital television broadcasting (DVB).
- Computer networks and multiplexed serial links.

Q3. What is Frequency Division Multiplexing (FDM)? What are its advantages, disadvantages and applications? Compare FDM and TDM. **[10 Marks]**

Frequency: ★★ HIGH (asked 4x in 7 papers)

Definition

Frequency Division Multiplexing (FDM) is an analogue multiplexing technique where the available bandwidth of a transmission medium is divided into multiple non-overlapping frequency bands. Each signal is modulated onto a different carrier frequency and all signals are transmitted simultaneously.

How FDM Works

Each input signal is modulated to a unique carrier frequency. Guard bands are inserted between adjacent channels to prevent interference. All modulated signals are combined and transmitted together. At the receiver, bandpass filters separate each channel by its carrier frequency.

Advantages

- Simple concept — all channels transmitted simultaneously.
- No time synchronisation required between sender and receiver.
- Well-suited to analogue signals (voice, broadcast).
- Reliable and proven technology.
- Channels are independent — failure in one does not affect others.

Disadvantages

- Guard bands waste some bandwidth.
- Crosstalk can occur if filters are not sharp enough.
- Intermodulation distortion when signals mix in non-linear amplifiers.
- Fixed allocation — inflexible when traffic varies.
- Not as efficient as TDM for digital data.

FDM vs TDM Comparison

Feature	FDM	TDM
Signal type	Analogue	Digital
Division	Frequency bands	Time slots
Transmission	All channels simultaneously	Channels take turns
Synchronisation	Not required	Required
Efficiency	Lower (guard bands waste BW)	Higher (ATDM)
Used in	Radio/TV, ADSL	GSM, T1/E1, ISDN

Applications of FDM

- AM and FM radio broadcasting.
- Traditional cable TV (analogue).
- ADSL broadband internet.
- First generation (1G) analogue cellular (AMPS).
- Telephone systems — frequency division of spectrum.

Q4. What is CDMA? How does it differ from GSM? [10 Marks]

Frequency: ★★ HIGH (asked 3x in 7 papers)

Definition of CDMA

Code Division Multiple Access (CDMA) is a channel access method used in mobile communications where multiple users share the same frequency band and transmit simultaneously, but each user's data is encoded with a unique spreading code (called a chip sequence). The receiver uses the same code to extract the intended signal.

How CDMA Works

- Each user is assigned a unique pseudo-random code (PN code).
- Data is multiplied (spread) with this code before transmission.
- The signal is spread over a wide frequency band.
- At the receiver, the same code is used to de-spread and extract the original signal.
- Other users' signals appear as background noise and are rejected.

Key Features of CDMA

- Uses spread spectrum technology (Direct Sequence Spread Spectrum — DSSS).
- Soft handover — can connect to multiple base stations simultaneously.
- Inherently more secure as spreading code is not easily detected.
- Supports variable data rates.

CDMA vs GSM Comparison

Feature	CDMA	GSM
Full form	Code Division Multiple Access	Global System for Mobile Communication
Multiple access	DSSS spread spectrum	TDMA+FDMA
SIM card	No SIM (device-based)	Uses SIM card
Security	Higher (spread spectrum)	Lower (cloning possible)
Roaming	Limited international	Worldwide roaming
Handover	Soft handover	Hard handover
Capacity	Higher	Lower
Used in	USA (Verizon), South Korea	India, Europe, worldwide

Chapter 2: Cellular Networks, GSM & GPRS

Cellular network concepts, GSM, and GPRS are among the top-tested topics. Questions appear almost every semester, often asking for features, architecture, and comparisons.

Q5. What is a Cellular Network? Under what conditions is frequency reuse possible by different cells in the network? **[10 Marks]**

Frequency: ★★ ★ **VERY HIGH (asked 5x in 7 papers)**

Definition

A cellular network is a radio network distributed over land through cells, where each cell is served by at least one fixed-location transceiver, known as a Base Transceiver Station (BTS) or cell site. Mobile phones connect to the network through the BTS with the strongest signal.

Architecture of Cellular Network

- Cell: The basic geographic unit. Each cell is represented as a hexagon in theory.
- Base Transceiver Station (BTS): Handles the radio interface to mobile devices.
- Base Station Controller (BSC): Controls multiple BTSs; manages radio resources.
- Mobile Switching Centre (MSC): Routes calls between cells and to PSTN.
- Home Location Register (HLR): Database of subscriber information.
- Visitor Location Register (VLR): Stores info about roaming subscribers.

How Cellular Networks Work

When a mobile device makes a call, it connects to the nearest BTS. As the user moves, the call is handed from one BTS to the next (handover). The MSC coordinates all switching. Multiple users share spectrum through FDMA, TDMA, or CDMA techniques.

Frequency Reuse

Frequency reuse is the concept of using the same radio frequencies in geographically separated cells. Because radio signals weaken with distance, the same frequency can be used again once cells are far enough apart.

Conditions for Frequency Reuse

- Distance Condition: The distance between co-channel cells (D) must be large enough that interference is tolerable. The ratio D/R (where R is cell radius) must exceed a threshold (typically $D/R \geq 4.6$ for $N=7$ cluster).
- Cluster Size (N): The number of cells in a cluster before the frequency pattern repeats. Common values: $N = 4, 7, 12$. Larger N means less interference but lower spectrum efficiency.
- Co-channel Interference Ratio: Signal-to-interference ratio (SIR) must be maintained above acceptable limits.
- Cell Size: Smaller cells allow more frequent reuse in a given area (microcells, picocells).
- Terrain and Obstacles: Buildings, hills can absorb signals — allowing closer reuse.
- Antenna Directivity: Directional antennas (sectorisation) reduce interference — each cell can be divided into 3 or 6 sectors.

Formula

Frequency Reuse Factor = $1/N$, where $N = i^2 + ij + j^2$ (i, j are non-negative integers). For $N=7$ ($i=2, j=1$): each frequency is reused once every 7 cells.

Q6. What is GSM technology? Explain its features, advantages, and disadvantages. Write a short note about cells used in GSM. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 4x in 7 papers)

What is GSM?

Global System for Mobile Communications (GSM) is a standard developed by the European Telecommunications Standards Institute (ETSI) for 2G digital cellular networks. It is the most widely used mobile communication standard globally.

GSM Architecture

- Mobile Station (MS): The handset + SIM card used by the subscriber.
- Base Station Subsystem (BSS): Includes BTS and BSC — handles radio communication.
- Network Switching Subsystem (NSS): MSC, HLR, VLR, AuC, EIR — handles switching and databases.
- Operation and Support Subsystem (OSS): Network management and monitoring.

Multiple Access in GSM

GSM uses a combination of FDMA and TDMA. The 900 MHz band is divided into 200 kHz channels (FDMA). Each channel is further divided into 8 time slots (TDMA) supporting 8 simultaneous users.

Cells Used in GSM

- Macro Cells: Radius 1–35 km. Used in rural areas, on highways. Large coverage.
- Micro Cells: Radius 200 m–1 km. Used in urban areas with high density.
- Pico Cells: Radius < 200 m. Indoor use — airports, malls, office buildings.
- Umbrella Cells: Cover multiple smaller cells to handle overflow and handover.

Features of GSM

- International roaming — SIM-based identity allows roaming globally.
- Digital encryption of voice and data — secure communication.
- SMS (Short Message Service) — text messaging.
- GPRS (2.5G) and EDGE (2.75G) extensions for data.
- SIM (Subscriber Identity Module) — portable subscriber identity.
- Operates in 850, 900, 1800, and 1900 MHz bands.

Advantages

- Widely deployed worldwide — excellent roaming support.
- Clear voice quality compared to 1G analogue.
- International standardisation — interoperability.
- SIM card allows easy device switching.
- Built-in security and authentication.

Disadvantages

- Limited data rates (9.6 kbps for basic GSM).
- TDMA creates pulsed interference that affects some electronics.
- Not as secure as CDMA (SIM cloning possible).
- Being replaced by 3G/4G/5G — ageing technology.

Q7. What is GPRS? Explain its features and the services offered by it. [10 Marks]

Frequency: ★★ HIGH (asked 4x in 7 papers)

What is GPRS?

General Packet Radio Service (GPRS) is a packet-switched technology that extends the GSM standard to enable mobile data communication. It is often referred to as 2.5G because it falls between 2G (GSM) and 3G in terms of capability. GPRS was standardised by ETSI in 1997.

Key Difference from GSM

GSM uses circuit switching — a dedicated channel is reserved for the entire duration of a call. GPRS uses packet switching — data is sent in packets and resources are used only when data is actually transmitted, making it much more efficient for internet usage.

GPRS Architecture Additions

- SGSN (Serving GPRS Support Node): Routes data packets between mobile and GGSN. Handles mobility and security.
- GGSN (Gateway GPRS Support Node): Connects GPRS network to external data networks (Internet, X.25).
- PCU (Packet Control Unit): Part of BSC; manages radio resource allocation for packet data.

Features of GPRS

- Data rates: 21.4 kbps to 171.2 kbps (theoretical maximum, multiple slots).
- Always-on connectivity — no need to dial up; billed per data volume not time.
- Packet-switched — efficient use of radio resources.
- Multiple time slots — a single user can use up to 8 time slots for data.
- IP-based data transport — supports TCP/IP and UDP/IP.
- WAP (Wireless Application Protocol) support for mobile web browsing.
- QoS classes: background, interactive, streaming, conversational.

Services Offered by GPRS

- Mobile Internet Browsing: Access websites via WAP browser.
- MMS (Multimedia Messaging Service): Send photos, audio, video via messages.
- Email: Send and receive email on mobile devices.
- Push-to-talk over cellular (PoC).
- Location-based services using GPS + GPRS.
- Mobile banking and m-commerce applications.
- Remote monitoring and telemetry.
- Instant messaging (early mobile IM services).

Chapter 3: 3G, 4G, 5G & LTE Networks

Questions on network generations (3G vs 4G, 4G features, 5G, LTE) appear every semester. Pay special attention to feature comparisons between generations.

Q8. What is a 4G Network? How does it differ from a 3G Network? What are its advantages and features? **[10 Marks]**

Frequency: ★★ HIGH (asked 4x in 7 papers)

What is 4G?

4G (Fourth Generation) is the fourth generation of mobile telecommunications technology. It follows 3G and provides ultra-broadband internet access through smartphones and other mobile devices. The standard is defined by the ITU-R and includes technologies like LTE (Long-Term Evolution) and WiMAX.

Key Features of 4G

- Data speeds: 100 Mbps (mobile) to 1 Gbps (stationary users).
- All-IP network — voice, video, and data all carried over IP (VoIP for calls).
- OFDM (Orthogonal Frequency Division Multiplexing) for air interface.
- MIMO (Multiple Input Multiple Output) antennas for higher throughput.
- Very low latency (~50 ms), enabling real-time applications.
- Supports HD video calling, online gaming, mobile TV.
- Advanced security — AES encryption.
- Inter-technology handover — seamless transition between 4G, 3G, WiFi.

4G vs 3G Comparison

Feature	3G	4G
Speed	2–21 Mbps	100 Mbps – 1 Gbps
Technology	WCDMA, CDMA2000	LTE, WiMAX
Latency	~100–500 ms	~50 ms
Frequency	850/1900/2100 MHz	700/1700/2100/2600 MHz
Network type	Circuit+Packet switched	All-IP, packet switched
Voice	Circuit switched	VoIP over IP
Services	Video calls, mobile internet	HD video, gaming, IoT

Advantages of 4G

- Ultra-fast data transfer for HD streaming.
- Lower latency improves real-time applications.
- Supports IoT devices, smart applications.
- Better spectral efficiency — more data per MHz.
- Enables high-quality VoIP calls.

Q9. Explain the features of 5G Network. Describe its advantages. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 3x in 7 papers)

What is 5G?

5G is the fifth generation of mobile network technology, succeeding 4G LTE. It was commercially launched in 2019. 5G uses three spectrum bands: low-band (sub-1 GHz), mid-band (1–6 GHz), and high-band/mmWave (24–100 GHz).

Key Features of 5G

- Speed: Peak download speed up to 20 Gbps; average 100 Mbps–1 Gbps.
- Ultra-low latency: As low as 1 ms (vs 50 ms in 4G). Enables real-time control.
- Massive connectivity: Up to 1 million devices per square kilometre.
- Network slicing: Virtualised sub-networks tailored for specific use cases.
- Beamforming: Antennas focus signal toward specific users rather than broadcasting.
- Massive MIMO: Up to 128 antennas at base station for higher capacity.
- Edge computing integration: Processing closer to devices reduces latency.
- Energy efficiency: 100x more energy efficient per bit than 4G.
- OFDM modulation with flexible numerology.

Advantages of 5G

- Enables autonomous vehicles (ultra-low latency control).
- Supports augmented and virtual reality applications.
- Enables smart cities, smart grid, and Industry 4.0.
- Better coverage in dense urban areas with small cells.
- Remote surgery and telemedicine (precision, low latency).
- Massive IoT deployments — millions of sensors in a city.
- High bandwidth streaming — 8K video, holograms.

Q10. What is LTE? Explain its features. **[10 Marks]**

Frequency: ★ **MEDIUM** (asked 2x in 7 papers)

What is LTE?

Long-Term Evolution (LTE) is a standard for high-speed wireless broadband communication for mobile devices. It is often marketed as 4G LTE and is the primary technology used by 4G networks worldwide. LTE was developed by the 3GPP (3rd Generation Partnership Project).

Features of LTE

- Peak download speed: 100 Mbps; Upload: 50 Mbps (LTE Release 8).
- Uses OFDMA (downlink) and SC-FDMA (uplink) for efficient spectrum use.
- Flat architecture: eNodeB (base stations) connect directly to core network — no BSC.
- Evolved Packet Core (EPC): All-IP core network — simplified architecture.
- MIMO support (2x2, 4x4) for multiple data streams.
- Flexible bandwidth: 1.4, 3, 5, 10, 15, 20 MHz channels.

- Supports FDD and TDD duplex modes.
- Evolved from HSPA/UMTS — backward compatible.
- LTE-Advanced (LTE-A): Carrier aggregation up to 100 MHz, speeds up to 1 Gbps.

Chapter 4: Mobile Computing Architecture & Design

Mobile computing architecture, design considerations, and client-server models are core theory topics that appear in most papers, often in Q1 or Q2.

Q11. Explain Mobile Computing Architecture with the help of a diagram. **[10 Marks]**

Frequency: ★★★ **VERY HIGH** (asked 5x in 7 papers)

What is Mobile Computing?

Mobile computing is a technology that allows data transmission, voice, and video communication via a computer or other wireless enabled device without being connected to a fixed physical link.

Three-Tier Mobile Computing Architecture

The standard mobile computing architecture consists of three tiers:

Tier 1 — Mobile Client (Presentation Layer)

- Mobile devices: smartphones, tablets, laptops, PDAs.
- Runs the user interface and presentation logic.
- Communicates with the middle tier via wireless network.
- Has limited resources — CPU, memory, battery.

Tier 2 — Application Server (Business Logic Layer)

- Processes client requests and contains business logic.
- Often includes a Web Server and Application Server.
- Handles session management, authentication.
- Acts as intermediary between client and database.

Tier 3 — Backend Server (Data Layer)

- Database servers storing persistent data.
- Legacy enterprise systems, ERP, etc.
- Accessible via wired internet.

Architecture Diagram (Text Representation)

[Mobile Device] ↔ [Wireless Network / Air Interface] ↔ [Base Station / Access Point] ↔
[Mobile Gateway] ↔ [Application Server] ↔ [Database Server]

Key Components

- Mobile Host (MH): The end user device.
- Mobile Support Station (MSS): Fixed base station managing the wireless cells.
- Fixed Host (FH): Traditional fixed computers connected to wired backbone.

- Wireless Channel: Radio frequency communication between MH and MSS.

Design Considerations

- Limited bandwidth — optimise data transmission.
- Intermittent connectivity — disconnected operation support.
- Battery constraints — minimise power consumption.
- Security — encryption over wireless links.
- Heterogeneity — support diverse device capabilities.

Q12. Explain the design considerations for Mobile Computing with a focus on operating system considerations. **[10 Marks]**

Frequency: ★★ HIGH (asked 3x in 7 papers)

Design Considerations for Mobile Computing

Designing mobile computing systems requires addressing unique challenges that do not exist in traditional wired computing:

1. Limited Resources

- CPU: Mobile processors are slower and less powerful than desktop CPUs.
- Memory (RAM): Typically 4–16 GB vs 16–64 GB for desktops.
- Storage: Limited, though cloud storage compensates.
- Battery: Finite energy — all design must minimise power drain.

2. Wireless Communication

- Bandwidth is limited and variable — design must handle low bandwidth.
- Signal loss, fading, and interference must be managed.
- Latency is higher than wired — critical for real-time applications.

3. Mobility Support

- Location awareness — GPS and network-based positioning.
- Handover support — seamless switching between base stations.
- Disconnected operation — apps must work offline and sync later.

4. Security Considerations

- Wireless networks are vulnerable to eavesdropping.
- Need end-to-end encryption (TLS/SSL).
- Device theft — remote wipe, device encryption.

5. Operating System Considerations

- Resource Management: Mobile OS must manage CPU, memory, and battery aggressively. Background process limits, app suspension.
- Power Management: OS implements dynamic voltage/frequency scaling, aggressive sleep states.
- Connectivity Management: OS manages transitions between WiFi, cellular, Bluetooth seamlessly.
- Security Model: Sandboxed applications — apps cannot access each other's data without permission.

- UI Optimisation: OS designed for touch input, small screens — gesture support, scalable UI.
- Update Mechanism: OTA (Over-The-Air) updates for OS and applications.
- Examples: Android (Linux kernel), iOS (XNU kernel) — both designed with these constraints in mind.

Q13. With reference to Smart Client Architecture, explain Thick Client and Thin Client Architecture. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 3x in 7 papers)

Smart Client Architecture

Smart Client Architecture refers to how the processing load is distributed between the client (mobile device) and the server. There are two primary approaches:

1. Thick Client Architecture (Fat Client)

In thick client architecture, most of the application processing occurs on the client device itself. The server mainly provides data storage and retrieval services.

Characteristics:

- Heavy application logic runs on the mobile device.
- Large local data storage on the device.
- Works well in offline/disconnected mode.
- Requires powerful device hardware.

Advantages:

- Fast response — processing is local.
- Works without internet connectivity.
- Rich user interface possible.
- Reduces server load.

Disadvantages:

- High device resource requirements.
- Updates must be pushed to all clients.
- Data consistency challenges when offline.
- Security risk if device is lost.

Examples: Native Android/iOS apps, desktop applications.

2. Thin Client Architecture

In thin client architecture, most processing occurs on the server. The client device merely displays the user interface and sends user inputs to the server.

Characteristics:

- Minimal processing on the client device.
- Relies heavily on server and network.
- Interface delivered via browser or lightweight client.

Advantages:

- Low hardware requirements on client side.

- Easy to maintain — updates only on server.
- Centralised data — easier security management.
- Device-agnostic — works on any capable browser.

Disadvantages:

- Requires constant network connectivity.
- Server becomes a bottleneck.
- Higher latency for user interactions.
- Limited rich UI capabilities.

Examples: Web-based applications, cloud apps accessed via mobile browser.

Comparison Summary

Feature	Thick Client	Thin Client
Processing	On client	On server
Offline support	Yes	No
Hardware need	High	Low
Maintenance	Complex	Simple
Examples	Native apps	Web apps

Chapter 5: TCP/UDP, Mobile IP & WAP

Networking protocols for mobile environments — TCP vs UDP, Mobile IP, WAP — are regularly tested especially in the context of mobility challenges.

Q14. Compare and contrast TCP and UDP in the context of mobile computing. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 3x in 7 papers)

TCP — Transmission Control Protocol

TCP is a connection-oriented, reliable transport layer protocol. It establishes a connection before data transfer and guarantees delivery, ordering, and error checking.

Key Features of TCP:

- Three-way handshake for connection establishment (SYN, SYN-ACK, ACK).
- Reliable delivery — acknowledgements and retransmission.
- Flow control — prevents sender from overwhelming receiver.
- Congestion control — backs off when network is congested.
- Ordered delivery — data arrives in sequence.
- Full duplex — simultaneous bidirectional communication.

UDP — User Datagram Protocol

UDP is a connectionless, unreliable transport layer protocol. It sends data without establishing a connection and does not guarantee delivery or ordering.

Key Features of UDP:

- No connection setup — datagrams sent immediately.
- No acknowledgements — lower overhead.
- No flow or congestion control.
- Lower latency — suitable for real-time applications.
- Best-effort delivery.

TCP vs UDP in Mobile Computing

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Guaranteed	Not guaranteed
Ordering	Ordered	Unordered
Speed	Slower (overhead)	Faster (no overhead)
Use case	File transfer, web, email	Video streaming, VoIP, gaming
Mobile issue	Slow on handover/packet loss	Better for real-time mobile

TCP Issues in Mobile Computing

- TCP's congestion control misinterprets wireless packet loss as network congestion — unnecessarily reduces speed.
- Handover causes temporary connection loss — TCP triggers unnecessary retransmissions.
- Solutions: Mobile TCP (M-TCP), Indirect TCP (I-TCP), Snooping TCP.

Q15. Explain the objective of Mobile IP. Explain the working of Mobile IP with the help of a diagram. **[10 Marks]**

Frequency: ★ **MEDIUM** (asked 2x in 7 papers)

What is Mobile IP?

Mobile IP (MIP) is an Internet Engineering Task Force (IETF) standard protocol (RFC 5944) that allows mobile device users to move from one network to another while maintaining a permanent IP address. Without Mobile IP, moving to a new network would require obtaining a new IP address and breaking existing connections.

Objectives of Mobile IP

- Maintain ongoing connections when moving between networks.
- Preserve IP address (Home Address) even when away from home network.
- Transparent to upper-layer applications — TCP sessions remain intact.
- Provide mobility without changing routing tables on the internet.

Key Entities in Mobile IP

- Mobile Node (MN): The mobile device with a permanent Home Address.

- Home Agent (HA): Router at the home network. Maintains location of MN. Tunnels packets to MN when away.
- Foreign Agent (FA): Router at the visited network. Provides services to MN while visiting.
- Care-of Address (CoA): Temporary IP address assigned to MN at foreign network.

Working of Mobile IP

1. MN moves from home network to foreign network.
2. FA advertises its presence. MN obtains a Care-of Address (CoA).
3. MN sends Registration Request to HA (via FA), containing CoA.
4. HA records the binding: Home Address ↔ CoA.
5. HA sends Registration Reply confirming binding.
6. Correspondent Node (CN) sends packets to MN's Home Address.
7. HA intercepts packets and tunnels them (IP-in-IP encapsulation) to the CoA.
8. FA receives the tunnelled packet, decapsulates and delivers to MN.
9. Return path: MN sends packets directly to CN (triangle routing).

Diagram (Text Form)

[CN] → [Home Network: HA] → (tunnel) → [Foreign Network: FA] → [MN]

[MN] → [CN] directly (return path — triangle routing)

Problems with Mobile IP

- Triangle routing inefficiency — return path via home network.
- Security risks — binding update vulnerabilities.
- Solution: Route Optimisation — MN sends Binding Update directly to CN.

Q16. What is WAP? How does it work? Discuss WAP Protocol Stack. **[10 Marks]**

Frequency: ★ **MEDIUM** (asked 2x in 7 papers)

What is WAP?

Wireless Application Protocol (WAP) is a technical standard for accessing information over a mobile wireless network. It was developed by the WAP Forum (later Open Mobile Alliance) in 1998 to allow mobile phones with small screens and limited bandwidth to access internet content.

How WAP Works

WAP works as a translation layer between mobile devices and internet servers. WAP content is written in WML (Wireless Markup Language), a compact version of HTML. A WAP Gateway translates HTTP/HTML content from web servers into WML format suitable for mobile devices.

WAP Protocol Stack

WAP defines a layered protocol stack, similar to OSI:

- WAE (Wireless Application Environment): Top layer. Contains WML browser, WMLScript. Application environment for mobile internet.
- WSP (Wireless Session Protocol): Session management. Binary encoding of HTTP for efficiency over wireless.
- WTP (Wireless Transaction Protocol): Transaction support — request/response. Replaces TCP for WAP.

- WTLS (Wireless Transport Layer Security): Security layer. Encryption, authentication, integrity. Mobile equivalent of TLS/SSL.
- WDP (Wireless Datagram Protocol): Adaptation layer that provides a consistent interface to upper layers regardless of underlying bearer network (GSM, GPRS, CDMA).
- Bearer Layer: Physical network — GSM, GPRS, CDMA, etc.

WAP Gateway

The WAP Gateway sits between the mobile device and the internet. It converts WAP requests into HTTP requests for web servers, and converts HTML responses into WML for mobile devices. It also handles WTLS-to-SSL/TLS conversion.

Limitations of WAP

- Limited compared to full HTML browsers.
 - WAP gap — decryption at gateway was a security risk.
 - Superseded by mobile browsers supporting full HTML5.
-

Chapter 6: Mobile Application Development

Mobile app development — platforms, IDEs, development phases, Android, iOS, J2ME, J2EE — is tested every semester across multiple questions.

Q17. Explain the features of an Integrated Development Platform (IDE) for Mobile Apps. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 4x in 7 papers)

What is an IDE for Mobile Apps?

An Integrated Development Environment (IDE) for mobile applications is a software suite that provides comprehensive tools for developing, testing, debugging, and deploying mobile applications on a single platform.

Key Features of a Mobile IDE

1. Code Editor

- Syntax highlighting for Java, Kotlin, Swift, Dart, etc.
- Code completion (IntelliSense) — auto-suggestion of methods and variables.
- Code refactoring support.
- Error highlighting in real-time.

2. Emulator / Simulator

- Virtual device emulator to test apps without physical hardware.
- Supports multiple screen sizes, OS versions.
- Simulate GPS, camera, sensors, network conditions.

3. Debugger

- Step-through debugging — execute code line by line.

- Breakpoint support.
- Variable inspection, memory profiling.
- Logcat (Android) / Console output for runtime logs.

4. Build System

- Automated compilation and packaging (APK for Android, IPA for iOS).
- Dependency management (Gradle, Maven, CocoaPods).
- Build variants (debug/release).

5. UI Designer

- Drag-and-drop layout editor (XML-based for Android).
- Preview across multiple device sizes.
- Constraint Layout, Material Design component library.

6. Version Control Integration

- Built-in Git support — commit, push, pull within IDE.
- Diff and merge tools.

7. Testing Tools

- Unit testing framework integration (JUnit, XCTest).
- UI testing automation (Espresso, XCUITest).
- Code coverage reports.

8. Deployment / Publishing

- Direct deployment to device or store submission tools.
- Google Play / App Store integration.

Popular Mobile IDEs

- Android Studio (by Google) — for Android development (Kotlin/Java).
- Xcode (by Apple) — for iOS/macOS development (Swift/Objective-C).
- Visual Studio Code + plugins — cross-platform.
- Flutter SDK / Dart IDE — for cross-platform apps.
- React Native (with VS Code) — JavaScript/TypeScript cross-platform.

Q18. Explain the features of iOS. Explain the architecture of iOS. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 4x in 7 papers)

What is iOS?

iOS is the mobile operating system developed by Apple Inc., first released in 2007 for the iPhone. It is closed-source and exclusive to Apple hardware (iPhone, iPad, iPod Touch). iOS is known for its security, smooth performance, and premium user experience.

Features of iOS

- Touch-based Interface: Multi-touch gestures — tap, swipe, pinch, rotate.
- Security: Secure Enclave hardware chip, Face ID/Touch ID biometric authentication, app sandboxing.
- App Store: Curated marketplace; strict review process ensures quality and safety.
- Siri: Built-in AI assistant with NLP capabilities.

- iCloud Integration: Seamless sync across all Apple devices.
- Continuity: Handoff between iPhone, iPad, Mac — start on one, continue on another.
- Privacy: App Tracking Transparency, Location permission granularity.
- AirDrop: Fast peer-to-peer file sharing.
- Swift/Objective-C: Powerful programming languages for app development.
- CarPlay and HealthKit: Integration with cars and health data platforms.

iOS Architecture Layers

iOS architecture is layered, from hardware abstraction to high-level frameworks:

Layer 1 — Core OS (Lowest)

- Darwin (XNU kernel — hybrid of Mach microkernel + BSD Unix).
- Low-level networking, file system, security.
- Accelerate Framework for math/signal processing.

Layer 2 — Core Services

- Foundation framework — strings, collections, networking.
- Core Data — object persistence and data management.
- Core Location — GPS and location services.
- CloudKit — iCloud data sync.

Layer 3 — Media

- Core Graphics, Core Animation, Core Image.
- AVFoundation — audio/video playback and recording.
- OpenGL ES / Metal — 3D graphics and GPU access.

Layer 4 — Cocoa Touch (Highest)

- UIKit — the main UI framework (buttons, views, controllers).
- MapKit — maps and location UI.
- EventKit — calendar and reminders.
- GameKit — multiplayer and leaderboards.

Q19. Draw and explain Android Architecture. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 3x in 7 papers)

What is Android?

Android is an open-source mobile operating system developed by Google, based on the Linux kernel. First released in 2008, it is the most widely used mobile OS in the world, used by over 70% of smartphones globally.

Android Architecture Layers (Bottom to Top)

1. Linux Kernel (Bottom Layer)

- Core of Android OS. Provides hardware abstraction.
- Handles memory management, process scheduling, security.
- Device drivers: camera, display, audio, Bluetooth, WiFi.
- Power management.

2. Hardware Abstraction Layer (HAL)

- Bridges the gap between hardware and Android framework.
- Provides standard interfaces that higher layers call.
- HAL modules for camera, audio, sensors, etc.

3. Android Runtime (ART) + Core Libraries

- ART (Android Runtime): Replaces Dalvik VM (Android 5.0+). Compiles bytecode to machine code using AOT (Ahead-of-Time) compilation.
- Core Libraries: Java core libraries providing standard Java APIs.

4. Native C/C++ Libraries

- WebKit — web browser engine.
- OpenGL ES — 3D graphics.
- SQLite — embedded relational database.
- Media framework — audio/video support.
- SSL — secure communications.

5. Application Framework

- Activity Manager — manages app lifecycle.
- Content Providers — share data between apps.
- Resource Manager — manages non-code resources.
- Notification Manager — system notification bar.
- View System — UI building blocks.

6. Applications (Top Layer)

- Home screen, phone dialler, contacts, browser, camera.
- Third-party apps from Google Play Store.
- All apps are written primarily in Java or Kotlin.

Architecture Diagram (Text Form)

[Applications] → [App Framework] → [Libraries + ART] → [HAL] → [Linux Kernel] → [Hardware]

Q20. Explain the features of J2ME. Explain main components of J2ME. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 3x in 7 papers)

What is J2ME?

Java 2 Micro Edition (J2ME) is a version of Java designed for resource-constrained devices such as mobile phones, PDAs, and set-top boxes. It was developed by Sun Microsystems (now Oracle) for devices with limited memory, display, and processing power.

Features of J2ME

- Platform independent — write once, run anywhere.
- Optimised for small devices — small footprint JVM.
- Security model — sandboxed execution.
- Networking support — HTTP, socket connections.
- GUI support via LCDUI (LCD User Interface).
- Multimedia support — audio, video playback.
- OTA (Over-the-Air) application delivery.

J2ME Architecture — Main Components

1. Configurations

Configurations define the minimum JVM and core libraries for a class of device:

- CDC (Connected Device Configuration): For devices with more resources — set-top boxes, PDAs. Supports full Java SE API subset.
- CLDC (Connected Limited Device Configuration): For highly constrained devices — basic mobile phones. Minimal Java libraries. Uses KVM (Kilobyte Virtual Machine).

2. Profiles

Profiles add domain-specific APIs on top of configurations:

- MIDP (Mobile Information Device Profile): Built on CLDC. For mobile phones and pagers. Includes LCDUI (GUI), RMS (Record Management System), networking, lifecycle management.
- PDAP (PDA Profile): For PDAs, built on CDC.

3. Optional Packages

- Bluetooth API, Wireless Messaging API (WMA), Mobile Media API (MMAPI).
- Location API, Web Services API.

MIDlet (Mobile Application)

A J2ME application is called a MIDlet. MIDlets have three states: Paused, Active, Destroyed. They use the MIDlet lifecycle (startApp, pauseApp, destroyApp methods).

Record Management System (RMS)

RMS is J2ME's persistent data storage mechanism. Data is stored in record stores (byte arrays). No SQL — records are accessed by record ID.

Chapter 7: Data Storage, Synchronisation & XML

Mobile data management — data store methods, synchronisation, XML parsing — appears in almost every paper. These are practical topics with consistent question patterns.

Q21. Explain Mobile Device Data Store Methods / Architecture patterns for mobile device data store. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 4x in 7 papers)

Introduction

Mobile devices require efficient data storage solutions that work within constraints of limited memory, intermittent connectivity, and battery life. Several patterns exist for mobile data storage.

1. On-Device Local Storage Methods

a) Shared Preferences / Key-Value Store

- Simple key-value pairs for configuration data, user preferences.
 - Android: SharedPreferences API. iOS: UserDefaults.
-

- Fast access, small data only.

b) File Storage

- Store data as flat files — text, binary, JSON, CSV.
- Internal storage (private) or External storage (SD card).
- Suitable for images, documents, log files.

c) SQLite Database

- Relational database embedded in mobile OS.
- Supports SQL queries — structured data storage.
- Android uses SQLite via Room library. iOS uses Core Data (which can use SQLite backend).
- Suitable for structured data: contacts, messages, records.

d) Object Storage / ORM

- Object-Relational Mapping tools — Realm Database, Room (Android).
- Store objects directly without writing SQL.

2. Remote / Cloud Storage

- Data stored on remote servers and accessed via API.
- Firebase Realtime Database, Firestore — cloud-sync in real time.
- REST APIs — app calls backend database server.

3. Architecture Patterns**Repository Pattern**

- Abstracts data source behind a repository interface.
- App logic doesn't know whether data comes from local DB or network.
- Enables offline-first: return local data, sync with server in background.

Cache-First Pattern

- Always check local cache first — serve cached data if available.
- Refresh cache from server in the background.
- Reduces network calls, improves performance.

Sync Pattern

- Periodic or real-time synchronisation between local and server data.
- Delta sync — only changed records transferred.
- Conflict resolution strategy needed.

Q22. What do you mean by XML Parsing? Explain with an example. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 3x in 7 papers)

What is XML?

XML (eXtensible Markup Language) is a markup language designed to store and transport data in a human-readable and machine-readable format. Unlike HTML which defines display, XML focuses purely on data structure.

What is XML Parsing?

XML Parsing is the process of reading an XML document and extracting its data elements (tags, attributes, text content) so they can be used by an application. A parser reads the XML file and creates a data structure (tree or event stream) in memory.

Types of XML Parsers

1. DOM Parser (Document Object Model)

- Loads the entire XML document into memory as a tree.
- Allows random access to any element.
- Higher memory usage — not suitable for very large files or mobile.
- Best for small XML files requiring multiple reads.

2. SAX Parser (Simple API for XML)

- Event-driven — triggers events (startElement, endElement, characters) as it reads.
- Does NOT load entire document into memory.
- Low memory usage — ideal for mobile devices.
- Sequential access only — cannot go back.

3. StAX Parser (Streaming API for XML)

- Pull-based event model — application pulls events as needed.
- More control than SAX, lower memory than DOM.

Example XML Document

```
<?xml version="1.0" encoding="UTF-8"?>
<students>
  <student id="101">
    <name>Rahul Sharma</name>
    <course>MCA</course>
    <marks>85</marks>
  </student>
  <student id="102">
    <name>Priya Singh</name>
    <course>MCA</course>
    <marks>92</marks>
  </student>
</students>
```

Parsing with DOM (Java Example)

```
DocumentBuilderFactory factory = DocumentBuilderFactory.newInstance();
DocumentBuilder builder = factory.newDocumentBuilder();
Document doc = builder.parse("students.xml");
NodeList list = doc.getElementsByTagName("student");
for (int i = 0; i < list.getLength(); i++) {
  Element student = (Element) list.item(i);
  String name = student.getElementsByTagName("name").item(0).getTextContent();
  System.out.println("Name: " + name);
}
```

Mobile XML Parsing

On Android, XML is widely used for layout files (res/layout/*.xml) and configuration. Android Studio uses a built-in XML parser. For data XML, the Android SDK provides DOM, SAX, and XmlPullParser (which is a StAX-like pull parser optimised for Android).

Q23. What is Data Synchronisation? Explain types of synchronisation methods between mobile client and remote server. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 3x in 7 papers)

What is Data Synchronisation?

Data synchronisation in mobile computing is the process of establishing consistency among data stored on a mobile device and data on a remote server, ensuring that any changes made on one are reflected on the other. This is critical because mobile devices often work offline.

Why Synchronisation is Needed

- Mobile devices may go offline — data changes must be tracked and merged later.
- Multiple devices may edit the same data — conflicts must be resolved.
- Users expect consistent data across devices and servers.

Synchronisation Methods

1. Full Synchronisation

- All data is transferred between client and server on every sync.
- Simple but inefficient — wastes bandwidth.
- Suitable for small datasets.

2. Incremental / Delta Synchronisation

- Only changes since last sync are transferred.
- Uses timestamps or change logs to identify modified records.
- Efficient — suitable for mobile environments.

3. Pull Synchronisation

- Client initiates the sync — requests data from server.
- Polling — client periodically checks for updates.
- Simple to implement; may introduce delay.

4. Push Synchronisation

- Server initiates — pushes updates to client when changes occur.
- Uses push notifications (FCM, APNs).
- Near real-time updates; requires persistent connection.

5. Bidirectional / Two-Way Sync

- Changes flow in both directions — client to server AND server to client.
- Requires conflict detection and resolution strategy.

6. SyncML Protocol

- Open standard for data synchronisation (OMA Data Synchronisation).
- XML-based sync protocol for contacts, calendar, tasks.

Conflict Resolution Strategies

- Server wins: Server data always overrides client data.

- Client wins: Client data always overrides server data.
- Last writer wins: Most recent timestamp wins.
- Manual resolution: User is asked to choose.

Chapter 8: Ad Hoc Networks, WLAN, WiMAX & Sensors

Wireless technologies beyond cellular — Ad Hoc networks, WLAN, WiMAX — and smart sensors/actuators are regularly tested, especially in Q1.

Q24. What are Ad Hoc Networks? Explain their features, advantages and disadvantages.
[10 Marks]

Frequency: ★★ HIGH (asked 4x in 7 papers)

What are Ad Hoc Networks?

An Ad Hoc network (also called Mobile Ad Hoc Network — MANET) is a decentralised, self-configuring, temporary wireless network of mobile devices that communicate directly with each other without relying on any pre-existing infrastructure (no base stations, no access points, no routers).

How Ad Hoc Networks Work

Each device in an ad hoc network acts as both a host and a router. Devices discover each other automatically. Data is routed through intermediate devices in a multi-hop manner if needed. The topology changes dynamically as devices join, leave, or move.

Types of Ad Hoc Networks

- MANET (Mobile Ad Hoc Network): General mobile ad hoc networks.
- VANET (Vehicular Ad Hoc Network): Between vehicles for traffic management.
- WSN (Wireless Sensor Network): Sensors forming an ad hoc network.
- FANET (Flying Ad Hoc Network): Between drones/UAVs.

Features

- Self-organising: No centralised administration — nodes configure themselves.
- Dynamic topology: Network structure changes as nodes move.
- Multi-hop routing: Data can pass through intermediate nodes.
- Decentralised: No single point of failure.
- Limited bandwidth: Radio communication constraints.
- Energy constrained: Battery-powered nodes.

Advantages

- No infrastructure needed — quick deployment in emergencies.
- Robust — no single point of failure.
- Low setup cost.
- Scalable — can add nodes easily.

- Useful in disaster relief, military operations, remote areas.

Disadvantages

- Security challenges — open medium, no central authority.
- Limited bandwidth and higher packet loss.
- Complex routing protocols needed (AODV, DSDV, DSR).
- Battery drain from routing overhead.
- Scalability issues with very large networks.
- Routing overhead increases with mobility.

Applications

- Military tactical communications.
- Disaster relief and emergency response.
- Sensor networks for environmental monitoring.
- Vehicle-to-vehicle communication (VANET).
- Conferences and temporary gatherings.

Q25. What are Smart Sensors? What are instances where mobile devices interact with their surroundings using sensors? **[10 Marks]**

Frequency: ★ **MEDIUM** (asked 2x in 7 papers)

What are Smart Sensors?

A smart sensor is a device that not only measures a physical quantity (temperature, light, pressure) but also processes the data locally, performs self-calibration, communicates data digitally, and may even make decisions. Unlike traditional passive sensors, smart sensors include processing capability (microcontroller/DSP) built-in.

Components of a Smart Sensor

- Transducer: Converts physical quantity to electrical signal.
- ADC (Analogue-to-Digital Converter): Digitises the signal.
- Microcontroller/DSP: Processes and analyses data.
- Memory: Stores calibration data and measurements.
- Communication Interface: I2C, SPI, Bluetooth, WiFi to send data.

Sensors in Mobile Devices

- Accelerometer: Detects acceleration/orientation. Used for screen rotation, step counting, shake gestures.
- Gyroscope: Measures angular velocity / rotation. Used in gaming, VR headsets, navigation.
- GPS (Global Positioning System): Location tracking. Used in maps, navigation, location-based services.
- Proximity Sensor: Detects nearby objects. Turns off screen when phone is held to ear during calls.
- Ambient Light Sensor: Measures surrounding light level. Auto-adjusts screen brightness.
- Magnetometer/Compass: Detects magnetic field for compass direction. Used in navigation apps.

- Barometer: Measures air pressure. Used for altitude detection, weather apps.
- Fingerprint Sensor: Biometric authentication.
- Camera: Computer vision — face recognition, AR, QR code scanning.
- Microphone: Voice commands, sound measurement.
- Heart Rate Sensor (wearables): Health monitoring.

Instances of Mobile Device Interaction with Surroundings

- Indoor Navigation: WiFi + accelerometer for indoor location when GPS is unavailable.
- Augmented Reality: Camera + accelerometer + gyroscope to overlay digital content on real world.
- Sleep tracking: Accelerometer detects movement patterns during sleep.
- Fall detection: Accelerometer + gyroscope detects sudden falls — alerts contacts.
- NFC Payments: NFC sensor enables contactless payment at POS terminals.
- Environmental monitoring: Temperature, humidity sensors in smart city deployments.

Q26. What are Actuators? Give some examples of actuators found in Robots. **[10 Marks]**

Frequency: ★ **MEDIUM** (asked 2x in 7 papers)

What are Actuators?

An actuator is a device that converts energy (electrical, hydraulic, pneumatic) into mechanical motion or action. Actuators are the 'output' components of a control system — they receive signals from a controller and perform a physical action (move, rotate, push, pull, vibrate).

Actuator vs Sensor

A sensor is an input device — it reads data from the environment. An actuator is an output device — it acts on the environment. Together they form sense-act loops in robotics and IoT systems.

Types of Actuators

- Electric Actuators: DC motors, stepper motors, servo motors — use electrical energy. Most common in robots.
- Hydraulic Actuators: Use pressurised fluid. Very high force. Used in heavy machinery.
- Pneumatic Actuators: Use compressed air. Fast response, moderate force. Used in factories.
- Thermal/Magnetic Actuators: Use temperature or magnetic fields for micro-scale movement.

Examples of Actuators in Robots

- DC Motor: Continuous rotation — used in robot wheels and drive systems. Simple speed and direction control.
- Servo Motor: Precise angular positioning. Used in robot arms, joints, pan-tilt heads. Controlled by PWM signals.
- Stepper Motor: Moves in discrete steps — precise control without feedback. Used in 3D printer axes, CNC machines.
- Linear Actuator: Converts rotational to linear motion. Used in robotic grippers, lift mechanisms.

- Solenoid: Electromagnetic coil — creates rapid linear push/pull motion. Used in robotic locks, valves.
- Piezoelectric Actuator: Uses piezo crystal deformation — very precise micro-movement. Used in surgical robots, optical systems.
- Vibration Motor (Haptic): Creates vibration feedback. Used in mobile phones and gaming controllers.
- Hydraulic Cylinder: Used in large industrial robots for heavy lifting.
- Shape Memory Alloy (SMA) Actuator: Changes shape with temperature. Used in soft robotics.

Chapter 9: Mobile OS, Handover & Mobile Cloud Computing

Mobile operating system functions, handover mechanisms, and mobile cloud computing are important theory topics tested with moderate frequency.

Q27. Explain the functions of a Mobile Operating System. **[10 Marks]**

Frequency: ★★ **HIGH** (asked 3x in 7 papers)

What is a Mobile OS?

A Mobile Operating System (Mobile OS) is system software that manages hardware and software resources of a mobile device, providing services to application programs. It acts as an interface between the hardware and the user.

Functions of a Mobile OS

1. Process Management

- Creates, schedules, and terminates processes/threads.
- Manages foreground and background processes.
- Suspends background apps to conserve battery.

2. Memory Management

- Allocates and deallocates memory to apps.
- Manages virtual memory and paging.
- Kills low-priority processes when memory is low.

3. File System Management

- Manages internal storage, SD card, USB storage.
- File system types: ext4 (Android), APFS (iOS).
- Manages permissions — each app has isolated storage.

4. Device Driver Management

- Manages hardware drivers — camera, display, WiFi, Bluetooth, GPS.
- Hardware Abstraction Layer (HAL) provides uniform interface.

5. Power Management

- Dynamic CPU frequency scaling (DVFS).
- Screen timeout, deep sleep modes.
- Battery usage monitoring and optimisation.
- Doze mode (Android) — severely limits background activity when idle.

6. Security

- Application sandboxing — apps isolated from each other.
- Permission model — apps request access to camera, location, etc.
- Encryption of storage and communications.
- Secure boot, verified OS integrity.

7. Networking

- Manages WiFi, Cellular (4G/5G), Bluetooth, NFC connections.
- Seamless handover between network types.
- VPN support.

8. User Interface

- Touch input processing — gesture recognition.
- Window management for multi-app display.
- Notification management.

9. App Lifecycle Management

- Manages app states: running, paused, stopped, destroyed.
- Handles app switching, recent apps.

Examples

- Android (Google): Based on Linux kernel. Open source. Largest market share.
- iOS (Apple): Based on XNU/Darwin. Closed source. Premium security.
- HarmonyOS (Huawei): Microkernel-based, distributed OS for all devices.

Q28. Elaborate the process of handing over control of a mobile device to a neighbouring cell. Discuss Soft and Hard Handover. **[10 Marks]**

Frequency: ★ **MEDIUM** (asked 2x in 7 papers)

What is Handover (Handoff)?

Handover (or handoff) is the process of transferring an active communication session (voice call or data connection) from one base station (cell) to another as the mobile user moves through the coverage area. The goal is to maintain uninterrupted service as the user moves.

When is Handover Triggered?

- Signal strength from current cell drops below a threshold.
- Signal strength from a neighbouring cell exceeds current cell.
- Current cell is overloaded — load balancing handover.
- User crosses cell boundary.

Handover Process (General Steps)

10. Mobile device monitors signal strength from all nearby base stations.
11. When signal from a neighbour consistently exceeds current BS (by hysteresis margin), mobile reports this to the network.
12. Network (MSC/RNC) decides to perform handover.
13. Resources (time slot, code, channel) are allocated at the target cell.
14. Mobile device switches to new base station.
15. Old connection is released.

Hard Handover (Break-Before-Make)

In hard handover, the connection to the old base station is broken before the connection to the new one is established. There is a brief interruption.

Characteristics:

- Used in GSM (TDMA/FDMA systems).
- Only one base station serves at a time.
- Brief call interruption during switch.
- Simpler to implement.
- Risk of dropped calls if new connection fails before old is released.

Soft Handover (Make-Before-Break)

In soft handover, the mobile device is simultaneously connected to two or more base stations during the handover period. The connection to the new BS is established before the old one is released.

Characteristics:

- Used in CDMA systems (WCDMA, CDMA2000).
- Multiple simultaneous BS connections using same frequency (spread spectrum).
- No interruption — seamless transition.
- Combines signals from multiple BSes (macro-diversity gain).
- Better at cell edges — improved call quality.
- More complex — requires more network resources.

Softer Handover

A special case of soft handover where the mobile connects to two sectors of the same base station (not different BSes). Used within sectorised cells in CDMA networks.

Q29. Describe the concept of Mobile Cloud Computing and its advantages for users. **[10 Marks]**

Frequency: ★ **MEDIUM** (asked 2x in 7 papers)

What is Mobile Cloud Computing (MCC)?

Mobile Cloud Computing (MCC) is a paradigm that integrates cloud computing into mobile environments. It allows mobile devices to offload computationally intensive tasks and storage to powerful cloud servers, accessed over a wireless network. This extends the capabilities of resource-constrained mobile devices.

Architecture of MCC

- Mobile Devices: Thin clients that initiate tasks and display results.
- Wireless Network: 4G/5G/WiFi connects devices to cloud.
- Internet: Routes traffic between mobile and cloud.
- Cloud Infrastructure: IaaS, PaaS, SaaS — virtual machines, storage, services.
- Mobile Cloud Server: Can include Mobile Edge Computing (MEC) — cloud closer to the user.

Advantages of MCC for Users

1. Extended Battery Life

- Computation offloaded to cloud — mobile processor does less work.
- Significant battery savings for intensive tasks like video rendering or AI.

2. Increased Storage

- Cloud provides virtually unlimited storage (Google Drive, iCloud, Dropbox).
- Not limited by device's physical storage.

3. Processing Power

- Access to cloud's powerful CPUs and GPUs.
- Enables complex tasks: AI inference, 3D rendering, big data analytics on a smartphone.

4. Data Synchronisation

- Automatic sync across devices — start on phone, continue on laptop.
- Data always up to date everywhere.

5. Cost Reduction

- No need for high-end expensive mobile hardware.
- Pay-per-use cloud services.

6. Reliability and Availability

- Cloud data backed up — no data loss if device is lost or damaged.
- High availability — 99.9%+ uptime for major cloud providers.

7. Scalability

- Cloud resources scale with demand — no need to buy more storage or compute.

Challenges of MCC

- Network dependency — requires stable internet connection.
- Latency — round trip to cloud adds delay.
- Privacy and security — sensitive data on third-party cloud.
- Bandwidth costs.

Chapter 10: Antennas, Signal Transmission & WLAN

Physical layer concepts — antennas, signal propagation, guided vs unguided transmission, and WLAN — are foundational topics that appear regularly.

Q30. What is an antenna? What is its functionality? On what factors does its size depend?
[10 Marks]

Frequency: ★★ HIGH (asked 3x in 7 papers)

What is an Antenna?

An antenna is a transducer that converts electrical signals into electromagnetic waves (for transmission) and electromagnetic waves back into electrical signals (for reception). It is the interface between an electronic circuit and the free-space propagation medium.

Functionality of an Antenna

Transmitting:

- Receives high-frequency alternating current from the transmitter.
- Converts it into electromagnetic waves that radiate outward through space.
- The radiation pattern determines the direction of signal transmission.

Receiving:

- Electromagnetic waves passing the antenna induce a small electrical current.
- This current is fed to the receiver and amplified.

Key Parameters of Antennas

- Gain: Measure of directivity and efficiency — how well it focuses energy in a direction.
- Radiation Pattern: 3D plot of signal strength vs direction. Omnidirectional vs directional.
- Bandwidth: Range of frequencies the antenna operates effectively on.
- Polarisation: Orientation of the electric field — vertical, horizontal, circular.
- VSWR (Voltage Standing Wave Ratio): Measure of impedance matching — lower is better.

Factors on Which Antenna Size Depends

1. Operating Frequency (Most Important)

The size of an antenna is primarily determined by the wavelength (λ) of the signal it is designed to transmit/receive. The relationship is:

$\lambda = c / f$ where $c = 3 \times 10^8$ m/s (speed of light), f = frequency

A half-wave dipole antenna has length = $\lambda/2$. At 900 MHz: $\lambda \approx 33$ cm $\rightarrow$ antenna length ≈ 16.5 cm.
 At 2.4 GHz (WiFi): $\lambda \approx 12.5$ cm $\rightarrow$ antenna ≈ 6.25 cm.

2. Gain Required

- Higher gain antennas (directional) tend to be physically larger.
- Yagi antennas: Long with multiple elements for high gain.

3. Bandwidth Requirements

- Wider bandwidth antennas require more complex and larger structures.

4. Physical Constraints

- In mobile devices — antennas must fit inside the handset.
- PIFA (Planar Inverted-F Antenna) used in phones — compact but efficient.

5. Environmental Factors

- In-body antennas (wearables) are smaller due to dielectric effect of body tissue.

Q31. What is guided transmission? How does it differ from unguided transmission? Give examples. **[10 Marks]**

Frequency: ★ **MEDIUM** (asked 2x in 7 papers)

Guided Transmission (Wired)

Guided transmission (also called bounded or wired transmission) uses a physical conductor or medium to guide electromagnetic signals from sender to receiver. The signal is confined to the medium.

Types of Guided Media

- **Twisted Pair Cable:** Two insulated copper wires twisted together. Used in telephone lines, Ethernet (Cat5, Cat6). Frequency: up to 600 MHz for Cat6. Cheap but susceptible to interference.
- **Coaxial Cable:** Central copper conductor surrounded by insulation, shield, and outer jacket. Used in cable TV, broadband internet. Less interference than twisted pair.
- **Optical Fibre:** Transmits light pulses. Glass or plastic core. Extremely high bandwidth (terabits/s). Immune to EMI. Used in backbone networks, long-haul links.

Unguided Transmission (Wireless)

Unguided transmission (also called unbounded or wireless transmission) uses electromagnetic waves through free space (air, vacuum) without any physical medium. The signal radiates in all directions (or a beam).

Types of Unguided Media

- **Radio Waves:** Frequency 3 kHz – 1 GHz. Omnidirectional. Penetrates walls. Used in AM/FM radio, cellular networks, WiFi.
- **Microwaves:** 1 GHz – 300 GHz. Directional (line-of-sight). Used in satellite links, point-to-point links, Bluetooth, WiFi (2.4/5 GHz).
- **Infrared:** 300 GHz – 400 THz. Short range, line-of-sight only. Used in TV remotes, old IrDA data transfer.

Guided vs Unguided Comparison

Feature	Guided	Unguided
Medium	Physical conductor	Free space
Mobility	Fixed	Mobile
Security	More secure	Less secure
Cost	Higher (infrastructure)	Lower
Interference	Lower	Higher
Speed	Higher (fibre)	Lower (wireless)

Quick Revision Guide — Exam Tips

Use this guide in the last 24 hours before your exam.

Top 10 Must-Prepare Topics

1. WDM — definition, advantages, disadvantages, applications (asked 6 times!)
2. Cellular Network + Frequency Reuse — conditions for reuse, cluster size formula.
3. Mobile Computing Architecture — 3-tier diagram, components.
4. GPRS — architecture additions (SGSN, GGSN), features, services.
5. 4G vs 3G comparison — speed, technology, latency, architecture table.
6. IDE features — 8 categories with examples (Android Studio, Xcode).
7. iOS Architecture — 4 layers with components.
8. Ad Hoc Networks — MANET, features, advantages, disadvantages, applications.
9. Mobile Data Store — 3 methods: Key-Value, SQLite, Remote/Cloud + patterns.
10. XML Parsing — DOM vs SAX vs StAX, Java code example.

Answer-Writing Strategy

- Start with a clear definition (2–3 lines). Examiners reward structured answers.
- Use labelled diagrams wherever possible — architecture questions need a diagram.
- Use bullet points for features/advantages/disadvantages — faster to write, easier to read.
- For comparison questions, always draw a table (5–6 rows minimum).
- End with applications or real-world examples — shows understanding.
- Q1 is compulsory (40 marks, 4 parts × 10 marks each) — never skip it.
- Attempt exactly 3 questions from Q2–Q5 as instructed.